1 Question (15 points)

The divine ratio of four numbers is calculated as

$$H = \frac{\frac{1}{p} - \frac{1}{q}}{\frac{1}{r} - \frac{1}{s}}.$$

Complete the following program which **calculates** and **prints** the divine ratio of the four integers p, q, r, and s. You may declare any necessary variables. You should not change any part of the given code.

```
public .......  Divine{
    public ....... ....... main(String[] args){

        int p=14, q=23, r=8, s=13;

        .
        .
        .
        .

        System.out.println("Divine ratio is equal to " + h);


    }
}
```

2 Question (15 points)

Write a main method and use for loops to calculate the following sum. **You are not allowed to use any if statements.**

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10}$$

3 Question (20 points)

Write a main method to produce the following output. You should use nested for loops or otherwise, you will not get any point.

```
12345
23451
34512
45123
51234
```

4 Question (20 points)

Write a structured and non-redundant program (including class and main method declaration) that will print the following figure. (You do not need to use for loops.)

```
\\""//
||||||
//""\\
||||||
\\""//

||||||
\\  //
//""\\
\\  //
||||||

//""\\
||||||
\\  //
//""\\
\\  //
```

5 Question (15 points)

The maximum distance that a car can travel depends on its fuel consumption, the number of passengers in the car and the number of liters of fuel in the fuel tank. Suppose that there is a car with 48 liters of fuel, 5 liters per 100 km fuel consumption (it consumes 5 liters of fuel in every 100 km with the driver inside only)) and 4 passengers. Every additional passenger increases the fuel consumption by %5.

Write a method to calculate the maximum distance that can be traveled by the car and print it.

Define at least 4 variables in your solution.

6 Question (15 points)

Assume that there are 30 days in a month and 365 days in a year. You are given an integer variable d that holds the total number of days. Write a program to convert d into days, month and year and print it. For example, 10000 days is equal to 27 years, 4 months and 25 days.